

Northwestern Medicine In-patient and Out-patient Beta-Lactam Cross-reactivity Chart for Drug Challenges

The risk of penicillin-cephalosporin cross-reactivity is lower than previously reported and predicted by (dis)similar beta-lactam side chains.

Clarification of **drug allergy history**, **clinical reasoning**, and **shared decision making** with the patient, when applicable, is advised when assessing the decision to drug challenge with a beta-lactam.

The NM beta-lactam cross-reactivity side chain chart is a guide and should be used with the NM beta-lactam Allergy Risk Assessment and Clinical Pathway for [Outpatients](#) or Non-Critically Ill Inpatients when assessing the decision to drug challenge with a beta-lactam.

Recommendations are based on published literature written by content experts that have considered similarities in structures, in vitro or skin test-based patterns of cross reactivity, challenge results, and clinical evidence, where available.

 ADSP Northwestern Medicine Antimicrobial and Diagnostic Stewardship Program		PENICILLINS					1 st GEN CEPHALOSPORINS			2 nd GEN CEPHALOSPORINS				3 rd GEN CEPHALOSPORINS						4 th	ADVANCED CEPHALOSPORINS			CARB	MONO
		Penicillin	Naf/Oxacillin Dicloxacillin	Amoxicillin	Ampicillin	Piperacillin	Cefadroxil	Cephalexin	<i>Cefazolin</i>	Cefaclor	Cefoxitin	Cefprozil	Cefuroxime	Cefdinir	Cefixime	Cefotaxime	Cefpodoxime	Ceftazidime	Ceftriaxone	Cefepime	Ceftaroline	Ceftolozane	Cefiderocol	Erta/Imipenem Meropenem	Aztreonam
PENICILLINS	Penicillin		▲	✖	✖	▲	▲	▲		▲		▲													
	Nafcillin/Oxacillin Dicloxacillin	▲		▲	▲	▲																			
	Amoxicillin (po)	✖	▲		✖	▲	✖	▲		▲		✖													
	Ampicillin	✖	▲	✖		▲	▲	✖		✖		▲													
	Piperacillin	▲	▲	▲	▲																				
1 ST GEN CEPHALOSPORINS	Cefadroxil (po)	▲		✖	▲			✖		▲		✖													
	Cephalexin (po)	▲		▲	✖		✖			✖		▲													
	<i>Cefazolin</i>																								
2 ND GEN CEPHALOSPORINS	Cefaclor (po)	▲		▲	✖		▲	✖				▲													
	Cefoxitin												✖												
	Cefprozil (po)	▲		✖	▲		✖	▲		▲															
	Cefuroxime (po)									✖						▲	▲	▲	▲	▲					
3 RD GEN CEPHALOSPORINS	Cefdinir (po)														✖										
	Cefixime (po)													✖											
	Cefotaxime											▲					✖	▲	✖	✖					
	Cefpodoxime											▲				✖		▲	✖	✖					
	Ceftazidime											▲				▲	▲		▲	▲			✖		✖
	Ceftriaxone											▲				✖	✖	▲	✖						
4 TH GEN CEPHALOSPORINS	Cefepime											▲				✖	✖	▲	✖						
ADVANCED CEPHALOSPORINS	Ceftaroline																								
	Ceftolozane																								
	Cefiderocol																	✖							✖
CARBAPENEMS	ertapenem, Meropenem, Imipenem																								
MONOBACTAMS	Aztreonam																	✖					✖		



= AVOID
Cross reaction known or
Identical R1 or R2 side chain



= CAUTION
Cross reaction less likely or
selective. Similar R1 ring or branch



= SUGGEST
Cross-reaction least likely
Dissimilar R1 or R2 side chain

CEFAZOLIN has a
unique side chain
with a very low
cross-reactivity
with penicillins
and
cephalosporins.

CEFAZOLIN may
be given without
prior testing in
patients with a
history of
anaphylaxis to a
penicillin

For more
information,
click [HERE](#)

For guidance in
use, click [HERE](#).

Khan DA, Banerji A, Blumenthal KG, et al. Drug allergy: A 2022 practice parameter update. *J Allergy Clin Immunol* 2022;150(6):1333-1393.
Zagursky RJ, Pichichero ME. Cross-reactivity in β -lactam Allergy. *J Allergy Clin Immunol Pract* 2018;6:72-81.
Romano A, et al. Cross-Reactivity and Tolerability of Cephalosporins in Patients with IgE-Mediated Hypersensitivity to Penicillins. *J Allergy Clin Immunol Pract* 2018;6(5):1662-72.
Romano Antonino. Immediate cephalosporin hypersensitivity: Allergy evaluation, skin testing, and cross reactivity with other beta-lactams. Update. August 2023